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Checkout devices, checkout assemblies, and checkout methods

Abstract

Implementations of this application provide a checkout device, a checkout assembly, and a checkout method. An example checkout device includes a housing, a display screen, and a Near Field Communication (NFC) module. The display screen is fastened to the housing, and is configured to display at least one of a to-be-paid amount of a product, payment method guidance, or a payment result. The NFC module is located in a space enclosed by the housing to be in close proximity to an electronic device to make an electronic payment. The display screen displays the payment method guidance to direct a consumer to quickly make an electronic payment. The display screen displays the to-be-paid amount of the product and the payment result.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of PCT Application No. PCT/CN2024/118727, filed on Sep. 13, 2024, which claims priority to Chinese Patent Application No. 202410172314.3, filed on Feb. 6, 2024, and each application is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

(1) This application relates to the field of electronic payment technologies, and in particular, to checkout devices, checkout assemblies, and checkout methods.

BACKGROUND

(2) With the rapid development of Internet technologies, electronic transactions have entered people's lives. However, an existing checkout device requires a merchant to hold a code scanner to scan a payment code of a user or requires a user to align a mobile terminal with a code scanning window of the checkout device, to complete a payment. This takes up time of the merchant and reduces checkout efficiency.

SUMMARY

(3) This application provides checkout devices, checkout assemblies, and checkout methods, to improve checkout efficiency.

(4) Some implementations of this application provide a checkout device, including: a housing; a display screen, fastened to the housing, where the display screen is configured to display a to-be-paid amount of a product and/or payment method guidance and/or a payment result; and an NFC module, located in a space enclosed by the housing, to be in close proximity to an electronic device to make an electronic payment.

(5) Further, the checkout device includes a scanning apparatus, and the scanning apparatus is

fastened to the housing for scanning the product to obtain a price of the product.

(6) Further, the scanning apparatus and the display screen are located on two opposite sides of the housing.

(7) Further, the checkout device includes an indicator light, the indicator light forms a ring, and light emitted by the indicator light surrounds the NFC module.

(8) Further, the checkout device is in a flat elongated shape, and includes a grip portion, the NFC module and the indicator light are located in a space enclosed by the grip portion, and the display screen is located above the grip portion in a length direction of the checkout device.

(9) Further, the checkout device includes a speaker located in the space enclosed by the housing, and a sound output hole is disposed on the housing for a sound emitted by the speaker to pass through.

(10) Some implementations of this application provide a checkout assembly, including a base and the above-mentioned checkout device. The checkout device is supported on the base.

(11) Further, the base includes a first magnetic component and a first conductive component, the checkout device includes a battery located in the housing, a second magnetic component, and a second conductive component fastened to the housing and exposed, the second conductive component is coupled to the battery, and when the checkout device is placed on the base, the first magnetic component and the second magnetic component attract and secure each other magnetically, and the first conductive component is in electrical contact with the second conductive component to charge the battery.

(12) Further, the base includes a first support portion, a connecting portion, and a second support portion, the first support portion is configured to be supported on a countertop, the connecting portion connects the first support portion and the second support portion, the second support portion is configured to support the checkout device, and the second support portion is disposed inclined relative to a vertical direction.

(13) Further, the second support portion includes a bottom plate portion and a first retaining edge and a pair of second retaining edges extending from the bottom plate portion, and when the checkout device is placed on the base, the first retaining edge abuts against a bottom end of the checkout device, and the pair of second retaining edges are respectively located on two opposite sides of the checkout device.

(14) Some implementations of this application provide an electronic payment method, including: A checkout device and an electronic device are provided, where the checkout device includes a display screen and an NFC module; the checkout device obtains a to-be-paid amount of a product, and displays the to-be-paid amount on the display screen; the checkout device provides payment method guidance; the electronic device approaches the NFC module based on the guidance, to make an electronic payment; and the display screen displays a payment result.

(15) Further, that the checkout device provides payment method guidance includes: The display screen provides visual guidance; and/or a speaker provides voice guidance, where the checkout device includes the speaker; and/or an indicator light provides light guidance, where the checkout device includes the indicator light.

(16) Further, that the display screen provides visual guidance includes: The picture includes a transition image that displays the to-be-paid amount; and that the display screen displays a payment result includes: After the electronic payment is completed, the display screen displays that the transition image moves toward the electronic device and disappears from the display screen.

(17) Further, that the checkout device obtains a to-be-paid amount of a product includes: The to-be-paid amount is manually entered on the display screen; and/or a scanning apparatus scans the product to obtain a price of the product, where the checkout device includes the scanning apparatus.

(18) In the implementations of this application, compared with code scanning payment, disposition of the NFC module in the checkout device makes it more convenient to make a payment, and

improves payment efficiency. The display screen is disposed in the checkout device, to display the to-be-paid amount of the product and/or the payment method guidance and/or the payment result. The payment method guidance can direct a consumer to quickly make an electronic payment, so as to improve payment efficiency. The to-be-paid amount of the product and the payment result are displayed. The consumer does not need to consult a merchant about the to-be-paid amount and whether the payment is completed. This frees the merchant, improves payment efficiency, and improves consumer experience.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a schematic diagram illustrating a checkout device, according to some example implementations of this application;
- (2) FIG. 2 is an exploded view illustrating the checkout device shown in FIG. 1, where only a part of the checkout device is shown;
- (3) FIG. 3 is a schematic diagram illustrating a top cover shown in FIG. 2;
- (4) FIG. 4 is a rear view illustrating the checkout device shown in FIG. 1;
- (5) FIG. 5 is an exploded view illustrating the checkout device shown in FIG. 4;
- (6) FIG. 6 is a schematic diagram illustrating a checkout assembly, according to some implementations of this application;
- (7) FIG. 7 is a schematic diagram illustrating a base of the checkout assembly shown in FIG. 6;
- (8) FIG. 8 is a side view illustrating the base shown in FIG. 7;
- (9) FIG. 9 is a bottom view illustrating the base shown in FIG. 7; and
- (10) FIG. 10 is a flowchart illustrating an electronic payment method according to some implementations of this application.

DESCRIPTION OF EMBODIMENTS

- (11) Example implementations are described in detail here, and examples of the example implementations are presented in the accompanying drawings. When the following descriptions relate to the accompanying drawings, unless specified otherwise, the same numbers in different accompanying drawings represent the same or similar elements. The implementations described in the following example implementations do not represent all implementations consistent with this application. On the contrary, the implementations are merely examples of apparatuses and methods consistent with some aspects of this application and described in detail in the appended claims.
- (12) User information (including but not limited to user equipment information, personal user information, etc.) and data (including but not limited to data used for analysis, stored data, displayed data, etc.) in this application are information and data that are authorized by a user or that are fully authorized by each party. Furthermore, related data need to be collected, used, and processed in compliance with relevant laws, regulations and standards of relevant countries and regions, and corresponding operation entries are provided for the user to choose to authorize or reject.
- (13) Terms used in this application are merely intended to describe specific implementations, but are not intended to limit this application. Terms “a”, “the”, and “this” in singular forms in this application and the appended claims are also intended to include plural forms, unless otherwise stated in the context clearly. It should be further understood that the term “and/or” used in this specification indicates and includes any or all possible combinations of one or more associated listed items.
- (14) It should be understood that although terms “first”, “second”, “third”, etc. may be used in this application to describe various types of information, the information should not be limited to the terms. These terms are merely used to distinguish between information of the same type. For

example, without departing from the scope of this application, first information can also be referred to as second information, and similarly, the second information can also be referred to as the first information. Depending on the context, the word “if” used here can be interpreted as “when” or “while” or “in response to determining”.

(15) As shown in FIG. 1 to FIG. 3, a checkout device **100** in some implementations of this application includes a housing **1**, a display screen **2**, and an NFC module **31**. The NFC module **31** is located in a space enclosed by the housing **1**, to be in close proximity to an electronic device to make an electronic payment. Compared with code scanning payment, it is more convenient to make a payment, and payment efficiency is improved. The electronic device includes but is not limited to a mobile phone, a tablet, etc.

(16) The display screen **2** is fastened to the housing **1**. When the checkout device **100** is used, the display screen **2** faces a consumer, and is configured to display a to-be-paid amount of a product and/or payment method guidance and/or a payment result. The payment method guidance can direct the consumer to quickly make an electronic payment, so as to improve payment efficiency. The to-be-paid amount of the product and the payment result are displayed. The consumer does not need to consult a merchant about the to-be-paid amount and whether the payment is completed. This frees the merchant, improves payment efficiency, and improves consumer experience.

(17) In some implementations, the payment method guidance displayed on the display screen **2** includes an image, text, an animation, etc.

(18) In some implementations, the display screen **2** provides visual guidance, and the picture includes a transition image that displays the to-be-paid amount. After the electronic payment is completed, the display screen displays that the transition image moves toward the electronic device and disappears from the display screen **2**, to achieve a similar effect of jumping from the display screen **2** to a payment device by the transition image, so as to improve payment experience.

(19) In some implementations, the transition image can be implemented through animation switching.

(20) In some implementations, the transition image can be spherical, to make the jump more vivid.

(21) In some implementations, the checkout device **100** includes an indicator light **32**, the indicator light **32** forms a ring, and light emitted by the indicator light **32** surrounds the NFC module **31**, to provide light guidance. This can attract attention of the consumer, to improve payment efficiency.

(22) In some implementations, when the checkout device **100** is in different states, the indicator light **32** can provide different guidance in different light forms. For example, in a normal standby state, the indicator light **32** can flash to attract attention of the consumer. When a payment is guided, the indicator light **32** can direct, by changing a color and by using a marquee, the consumer to make an electronic payment. A payment success and a payment failure can be displayed in different colors, to notify the payment result in a light form. The light form is not limited in this application.

(23) In some implementations, the checkout device **100** is a handheld checkout device, is in an overall flat elongated shape, and includes a grip portion **3**, the NFC module **31** and the indicator light **32** are located in a space enclosed by the grip portion **3**, and the display screen **2** is located above the grip portion **3** in a length direction of the checkout device **100**. The checkout device **100** is lightweight and easy to hold.

(24) In some implementations, the housing **1** includes a main body housing **13** and a circular top cover **14** fastened to the main body housing **13**, the NFC module **31** and the indicator light **32** are located below the top cover **14**, and the top cover **14** is light-transmissive for light from the indicator light **32** to transmit. The top cover **14** is provided with guiding text **141**, to guide a payment method. For example, the guiding text **141** is an NFC identifier, etc.

(25) In some implementations, the guiding text **141** is light-transmissive for the light emitted by the indicator light **32** to transmit. When the indicator light **32** guides a payment, the guiding text **141** is lit.

(26) In some implementations, the checkout device **100** includes a circuit board **34**, the indicator light **32** is a plurality of LED beads fastened to the circuit board **34**, and the plurality of LED beads are disposed in a circumferential direction of the circuit board **34**. In some implementations, the checkout device **100** includes a light guiding ring **35** covering the LED beads.

(27) In some implementations, the NFC module **31** is fastened to the top cover **14**.

(28) In some implementations, the main body housing **13** is provided with a cavity **130**, the circuit board **34** is located in the cavity **130**, the checkout device **100** includes a middle frame **36** and a transparent decorative cover **37**, the middle frame **36** is fastened to the main body housing **13**, the middle frame **36**, the top cover **14**, and the decorative cover **37** are fastened together, and the decorative cover **37** covers the top cover **14**. The middle frame **36** is disposed so that the top cover **14** protrudes from an upper surface of the main body housing **13**, to facilitate a grip.

(29) In some implementations, after the to-be-paid amount is manually entered on the display screen **2**, the display screen **2** can display the to-be-paid amount of the product and/or the payment method guidance and/or the payment result.

(30) As shown in FIG. **4** and FIG. **5**, in some implementations, the checkout device **100** includes a scanning apparatus **4**, and the scanning apparatus **4** is fastened to the housing **1** for scanning the product to obtain a price of the product. For example, the price of the product can be obtained by scanning a two-dimensional code of the product. In this application, a scanning method is not limited, provided that the price of the product can be obtained by scanning the product.

(31) In some implementations, the scanning apparatus **4** and the display screen **2** are located on two opposite sides of the housing **1**. In FIG. **1**, the display screen **2** is located on a front of the checkout device **100**, the scanning apparatus **4** is located on a back of the checkout device **100**, and the electronic device approaches the NFC module **31** from the front of the checkout device **100**, to make an electronic payment.

(32) In some implementations, the checkout device **100** includes a speaker **5** located in the space enclosed by the housing **1**. A sound output hole **131** is disposed on the housing **1** for a sound emitted by the speaker **5** to pass through. The speaker **5** can broadcast advertising information, the payment method guidance, the to-be-paid amount, payment completion, etc., can be more convenient to use, and can attract the consumer and improve checkout efficiency.

(33) In some implementations, the sound output hole **131** is disposed on the main body housing **13**.

(34) In some implementations, the checkout device **100** includes a main board **61** and a battery **62** located in the space enclosed by the housing **1**. The NFC module **31**, the circuit board **34**, the scanning apparatus **4**, the speaker **5**, and the battery **62** are coupled to the main board **61**.

(35) The housing **1** includes a rear cover plate **15**, the rear cover plate **15** is fastened to the main body housing **13**, the main board **61** and the battery **62** are located in a space enclosed by the rear cover plate **15** and the main body housing **13**, and a through hole **151** is disposed on the rear cover plate **15**, to expose the scanning apparatus **4**.

(36) In some implementations, the checkout device **100** includes a charging port (not shown in the figure) coupled to the battery **62**, to charge the battery **62**. In some implementations, the battery **62** can be replaced. In some implementations, the battery **62** may not be disposed, and the checkout device **100** supplies power through an external power supply.

(37) Optionally, the charging port can be of Type, etc.

(38) The checkout device **100** includes a plurality of keys **63**, and the keys **63** are disposed on a side edge of the checkout device **100**, and can include a power on/off key, a volume adjustment key, etc.

(39) As shown in FIG. **6** to FIG. **9**, some implementations of this application provide a checkout assembly, including a base **200** and the above-mentioned checkout device **100**. The checkout device **100** is supported on the base **200**.

(40) In some implementations, the checkout device **100** can be charged through the base **200**.

(41) In some implementations, the base **200** includes a first conductive component **71**. As shown in

FIG. 4, the checkout device **100** includes a second conductive component **64** fastened to the housing **1** and exposed, and the second conductive component **64** is coupled to the battery **62**. When the checkout device **100** is placed on the base **200**, the first conductive component **71** is in electrical contact with the second conductive component **64** to charge the battery **62**.

(42) In some implementations, the base **200** includes a first magnetic component, and the checkout device **100** includes a second magnetic component located in the housing **1**. When the checkout device **100** is placed on the base **200**, the first magnetic component and the second magnetic component attract and secure each other magnetically, to improve stability between the checkout device **100** and the base **200**, and ensure that the first conductive component **71** is in electrical contact with the second conductive component **64**. In addition, alignment between the first conductive component **71** and the second conductive component **64** does not need to be considered, and there is more convenience in use.

(43) In some implementations, the base **200** includes a first support portion **73**, a connecting portion **74**, and a second support portion **75**. The first support portion **73** is configured to be supported on a countertop, the connecting portion **74** connects the first support portion **73** and the second support portion **75**, the second support portion **75** is configured to support the checkout device **100**, and the second support portion **75** is disposed inclined relative to a vertical direction. When the checkout device **100** is placed on the base **200**, the checkout device **100** is in an inclined state, to help a consumer view the display screen **2** and facilitate a contactless payment between an electronic device and the NFC module **31**.

(44) In some implementations, the second support portion **75** includes a bottom plate portion **751** and a first retaining edge **752** and a pair of second retaining edges **753** extending from the bottom plate portion **751**. When the checkout device **100** is placed on the base **200**, the first retaining edge **752** abuts against a bottom end of the checkout device **100**, and the pair of second retaining edges **753** are respectively located on two opposite sides of the checkout device **100**, to improve stability of the checkout device **100**, so as to prevent the checkout device **100** from falling from the base **200**. In addition, the pair of second retaining edges **753** can further provide guidance, to direct the checkout device **100** to be assembled on the base **200**.

(45) The first retaining edge **752** can be connected to the pair of second retaining edges **753** to form a U shape, or may not be connected.

(46) When the checkout device **100** is not used, the checkout device **100** is placed on the base **200**, and can be charged or play an advertisement through the display screen **2**. When a payment needs to be collected, the checkout device **100** is removed from the base **200**, to manually enter an amount or use the scanning apparatus **4** to scan a product. After the amount is entered or scanning ends, the checkout device **100** is placed on the base **200** again for the consumer to view the payment amount and payment method guidance and bring the electronic device close to the NFC module **31** to make an electronic payment.

(47) As shown in FIG. 10, some implementations of this application provide an electronic payment method, including step **S1001** to step **S1005**.

(48) **S1001**: A checkout device **100** and an electronic device are provided, where the checkout device includes a display screen **2** and an NFC module **31**.

(49) **S1002**: The checkout device **100** obtains a to-be-paid amount of a product, and displays the to-be-paid amount on the display screen **2**.

(50) **S1003**: The checkout device **100** provides payment method guidance.

(51) **S1004**: The electronic device approaches the NFC module **31** based on the guidance, to make an electronic payment.

(52) **S1005**: The display screen **2** displays a payment result.

(53) In some implementations, that the checkout device provides payment method guidance includes: The display screen **2** provides visual guidance; and/or a speaker **5** provides voice guidance, where the checkout device includes the speaker **5**; and/or an indicator light **32** provides

light guidance, where the checkout device includes the indicator light **32**.

(54) In some implementations, that the display screen provides visual guidance includes: The picture includes a transition image that displays the to-be-paid amount; and that the display screen **2** displays a payment result includes: After the electronic payment is completed, the display screen displays that the transition image moves toward the electronic device and disappears from the display screen **2**.

(55) In some implementations, that the checkout device obtains a to-be-paid amount of a product includes: The to-be-paid amount is manually entered on the display screen **2**; and/or a scanning apparatus **4** scans a two-dimensional code of the product to obtain a price of the product, where the checkout device includes the scanning apparatus **4**.

(56) A person skilled in the art can easily think of another implementation solution of the present application after considering the specification and practicing the invention disclosed here. This application is intended to cover any variations, uses, or adaptations of this application. These variations, uses, or adaptations follow the general principles of this application and include common knowledge or conventional technical means in the art that are not disclosed in this application. The specification and the implementations are considered as examples, and the true scope and spirit of this application are indicated by the following claims.

(57) It is worthwhile to further note that the terms “include”, “comprise”, or any other variants thereof are intended to cover a non-exclusive inclusion so that a process, a method, a product, or a device that includes a list of elements not only includes those elements but also includes other elements which are not expressly listed, or further includes elements inherent to such process, method, product, or device. Without more constraints, an element preceded by “includes a . . . ” does not preclude the presence of additional identical elements in the process, method, product, or device that includes the element.

(58) The above-mentioned descriptions are merely example implementations of this application, but are not intended to limit this application. Any modification, equivalent replacement, improvement, etc. made without departing from the spirit and principle of this application should fall within the protection scope of this application.

Claims

1. A checkout device, comprising: a housing, wherein the housing comprises a grip portion that at least partially encloses a cavity space in the housing, the cavity space being at least partially covered by a top cover that protrudes from an upper surface of the housing; a display screen that is fastened to the housing and that is located above the grip portion in a length direction of the checkout device, wherein the display screen is configured to display at least one of a to-be-paid amount of a product, a payment method guidance, or a payment result; and a Near Field Communication (NFC) module located in the cavity space that is at least partially enclosed by the grip portion of the housing, to be in proximity to an electronic device to make an electronic payment.
2. The checkout device according to claim 1, wherein the checkout device comprises a scanning apparatus, and the scanning apparatus is fastened to the housing for scanning the product to obtain a price of the product.
3. The checkout device according to claim 2, wherein the scanning apparatus and the display screen are located on two opposite sides of the housing.
4. The checkout device according to claim 1, wherein the checkout device comprises an indicator light, the indicator light forms a ring, and light emitted by the indicator light surrounds the NFC module.
5. The checkout device according to claim 4, wherein the checkout device is in a flat elongated shape.

6. The checkout device according to claim 4, wherein the top cover is light-transmissive for light from the indicator light to transmit.
7. The checkout device according to claim 1, wherein the top cover has a circular shape.
8. The checkout device according to claim 1, wherein the cavity space is at least partially covered by a transparent cover that covers the top cover.
9. The checkout device according to claim 1, wherein the checkout device comprises a speaker located in a space enclosed by the housing, and a sound output hole is disposed on the housing for a sound emitted by the speaker to pass through.
10. A checkout assembly, comprising a base and a checkout device supported on the base, wherein the checkout device comprises: a housing, wherein the housing comprises a grip portion that at least partially encloses a cavity space in the housing, the cavity space being at least partially covered by a top cover that protrudes from an upper surface of the housing; a display screen that is fastened to the housing and that is located above the grip portion in a length direction of the checkout device, wherein the display screen is configured to display at least one of a to-be-paid amount of a product, a payment method guidance, or a payment result; and a Near Field Communication (NFC) module located in the cavity space that is at least partially enclosed by the grip portion of the housing, to be in proximity to an electronic device to make an electronic payment.
11. The checkout assembly according to claim 10, wherein the checkout device comprises a scanning apparatus, and the scanning apparatus is fastened to the housing for scanning the product to obtain a price of the product.
12. The checkout assembly according to claim 11, wherein the scanning apparatus and the display screen are located on two opposite sides of the housing.
13. The checkout assembly according to claim 10, wherein the checkout device comprises an indicator light, the indicator light forms a ring, and light emitted by the indicator light surrounds the NFC module.
14. The checkout assembly according to claim 13, wherein the checkout device is in a flat elongated shape.
15. The checkout assembly according to claim 10, wherein the checkout device comprises a speaker located in a space enclosed by the housing, and a sound output hole is disposed on the housing for a sound emitted by the speaker to pass through.
16. The checkout assembly according to claim 10, wherein the base comprises a first magnetic component and a first conductive component, the checkout device comprises a battery located in the housing, a second magnetic component, and a second conductive component fastened to the housing and exposed, the second conductive component is coupled to the battery, and when the checkout device is placed on the base, the first magnetic component and the second magnetic component attract and secure each other magnetically, and the first conductive component is in electrical contact with the second conductive component to charge the battery.
17. The checkout assembly according to claim 10, wherein the base comprises a first support portion, a connecting portion, and a second support portion, the first support portion is configured to be supported on a countertop, the connecting portion connects the first support portion and the second support portion, the second support portion is configured to support the checkout device, and the second support portion is disposed inclined relative to a vertical direction.
18. The checkout assembly according to claim 17, wherein the second support portion comprises a bottom plate portion and a first retaining edge and a pair of second retaining edges extending from the bottom plate portion, and when the checkout device is placed on the base, the first retaining edge abuts against a bottom end of the checkout device, and the pair of second retaining edges are respectively located on two opposite sides of the checkout device.
19. A method, comprising: providing a checkout device and an electronic device, wherein the checkout device comprises a housing, a display screen, and a Near Field Communication (NFC)

module, wherein the housing comprises a grip portion that at least partially encloses a cavity space in the housing, the cavity space being at least partially covered by a top cover that protrudes from an upper surface of the housing, and wherein the NFC module is located in the cavity space that is at least partially enclosed by the grip portion of the housing; obtaining, by the checkout device, a to-be-paid amount of a product; displaying, by the checkout device, the to-be-paid amount on the display screen that is fastened to the housing and that is located above the grip portion in a length direction of the checkout device; providing, by the checkout device, payment method guidance; making, by the electronic device, an electronic payment using the NFC module based on the guidance; and displaying, by the checkout device, a payment result on the display screen.

20. The method according to claim 19, wherein the providing, by the checkout device, payment method guidance comprises at least one of: providing, by the display screen, visual guidance; providing, by a speaker, voice guidance, wherein the checkout device comprises the speaker; or providing, by an indicator light, light guidance, wherein the checkout device comprises the indicator light.

21. The method according to claim 20, wherein the providing, by the display screen, visual guidance comprises: providing a transition image that displays the to-be-paid amount; and the displaying, by the display screen, a payment result comprises: after the electronic payment is completed, displaying, by the display screen, that the transition image moves toward the electronic device and disappears from the display screen.

22. The method according to claim 19, wherein the obtaining, by the checkout device, a to-be-paid amount of a product comprises at least one of: obtaining manually entered the to-be-paid amount on the display screen; or scanning, by a scanning apparatus, the product to obtain a price of the product, wherein the checkout device comprises the scanning apparatus.
